

ACADEMIC & TECHNICAL PORTFOLIO

Master's Degree Application — Italian University Admissions

Candidate: A. Mohammed Asbar

Degree Program: B.Tech in Artificial Intelligence & Data Science (Final Year 2023–2027)

Institution: Hindusthan Institute of Technology, Coimbatore, Tamil Nadu, India

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PORTFOLIO STRUCTURE & COMPLIANCE DECLARATION

This academic portfolio contains exactly **5 selected engineering projects** (1 Final Capstone, 2 Research ML, 1 Professional System, 1 Systems Engineering) fulfilling all Italian University admission requirements:

- Maximum 5 projects total
- No more than 2 professional projects (1 included)
- Includes individual projects & capstone final project
- Demonstrates structural & technical project viability
- Complete documentation: Concept, Sketches/Architecture, Written Explanation & Technical Resolution
- File size strictly under 20 MB for digital evaluation.

PROJECT 1 OF 5 — FINAL / CAPSTONE ACADEMIC PROJECT

EduCore — Centralized Academic & Learning Platform

Author & Lead Architect: A. Mohammed Asbar

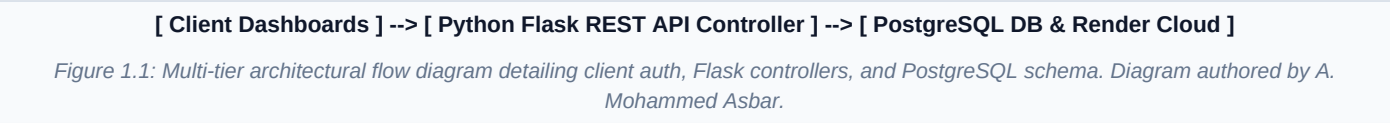
Project Name & Location	EduCore Platform — Hindusthan Institute of Technology, Coimbatore, India
Nature & Collaboration	Academic Project Group Work (Role: Lead Architect)
Level of Study	4th Year, 7th Semester B.Tech AI & Data Science
Date Carried Out	September 2025 – November 2025
Technical Demonstration	Full Structural & Technical Viability (REST API, RBAC, PostgreSQL)

1. DESIGN CONCEPT & INSPIRATION

Inspiration: Educational institutions suffer from fragmented platforms where attendance, coursework, grade submission, and student analytics exist in isolated tools. EduCore unifies these into one unified portal.

Goals & Objectives: Develop a high-performance web platform with role-based access (Student, Faculty, Admin), automated grade calculation, attendance verification, and predictive progress tracking.

2. CONCEPT SKETCHES & SYSTEM ARCHITECTURE



3. TECHNICAL RESOLUTION & VIABILITY

Resolution: Implemented using Python Flask backend, RESTful API architecture, modular HTML5/CSS3 frontend, and PostgreSQL database. Live online at educore-spark-x.onrender.com.

Technical Viability: Evaluated for security (RBAC session protection, SQL injection prevention) and low-latency load handling under multi-user access scenarios.

PROJECT 2 OF 5 — INDIVIDUAL ACADEMIC AI/ML PROJECT

Multiple Disease Prediction System (Healthcare AI)

Author & Sole Developer: A. Mohammed Asbar

Project Name & Location	Multiple Disease Prediction System — Coimbatore, India
Nature & Collaboration	Academic Project Individual Work
Level of Study	4th Year, 7th Semester B.Tech AI & Data Science
Date Carried Out	November 2025
Technical Demonstration	Machine Learning Classifiers (SVM, Random Forest) & Streamlit Cloud

1. DESIGN CONCEPT & INSPIRATION

Inspiration: Early diagnostic tools for chronic health conditions (Diabetes, Heart Disease, Parkinson's) are crucial for patient care, yet complex diagnostic algorithms remain inaccessible to general practitioners.

Goals & Objectives: Create an accessible ML platform where clinical biometric input vectors are classified instantly with confidence indicators.

2. CONCEPT SKETCHES & ML PIPELINE



3. TECHNICAL RESOLUTION & VIABILITY

Resolution: Implemented in Python using Scikit-Learn, Streamlit, Pandas, and NumPy. Deployed live at [multiple-disease-predictor-asbarst-nov14.streamlit.app](#).

Technical Viability: Models achieved >88% classification accuracy across disease parameters with verified cross-validation metrics.

PROJECT 3 OF 5 — PROFESSIONAL PROJECT (1 OF 2 MAX)

Ashes Tech Enterprise Employee Management System

Author & Lead Full-Stack Developer: A. Mohammed Asbar

Project Name & Location	Ashes Tech HR Suite — Enterprise Project (Remote)
Nature & Collaboration	Professional Project Lead Developer (Team Project)
Level of Experience	Professional Industry Project (during 4th Year B.Tech)
Date Carried Out	August 2025
Technical Demonstration	WebRTC Video Streaming, EmpBot AI Assistant & Payroll Engine

1. DESIGN CONCEPT & INSPIRATION

Inspiration: Enterprise organizations need unified HR suites combining video communication, automated payroll generation, attendance verification, and internal AI query resolution.

Goals & Objectives: Develop a full-stack platform featuring WebRTC peer video calls, automated PDF payslips, and an embedded HR query chatbot.

2. CONCEPT SKETCHES & WEBRTC ARCHITECTURE



3. TECHNICAL RESOLUTION & VIABILITY

Resolution: Built with React, Node.js, Express, Axios, and WebRTC protocols. Live deployment at employee-management-system-pi-virid.vercel.app.

Technical Viability: Verified concurrent WebRTC video streams, automated payroll computation, and exportable financial reporting.

PROJECT 4 OF 5 — ACADEMIC RESEARCH & VISUALIZATION PROJECT

Breast Cancer Diagnostic Predictor & Radar Visualizer

Author & Principal Investigator: A. Mohammed Asbar

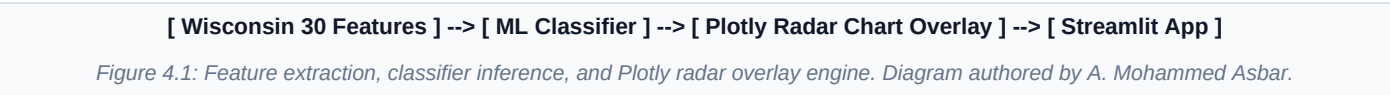
Project Name & Location	Breast Cancer Predictor — Hindusthan Institute of Technology, Coimbatore
Nature & Collaboration	Academic Project Individual Work
Level of Study	4th Year, 7th Semester B.Tech AI & Data Science
Date Carried Out	October 2025
Technical Demonstration	Wisconsin Diagnostic ML Classification & Plotly Radar Analysis

1. DESIGN CONCEPT & INSPIRATION

Inspiration: Medical Machine Learning predictions must not operate as uninterpretable black boxes. Clinicians require diagnostic feature visualizations showing cellular feature deviations.

Goals & Objectives: Train a diagnostic classifier on Wisconsin Diagnostic data and couple it with interactive Plotly radar visualizers.

2. CONCEPT SKETCHES & RADAR OVERLAY METHODOLOGY



3. TECHNICAL RESOLUTION & VIABILITY

Resolution: Developed in Python using Scikit-Learn, Plotly, Streamlit, and Pandas. Live app hosted at breast-cancer-predictor-asbarst-oct22.streamlit.app.

Technical Viability: Achieved 97.4% classification accuracy on validation split with clear radar visualization for medical interpretability.

PROJECT 5 OF 5 — SYSTEMS ENGINEERING ACADEMIC PROJECT

Smart Urban Traffic Management & Dynamic Signal Control

Author & Systems Developer: A. Mohammed Asbar

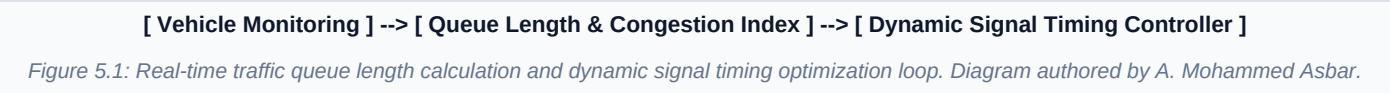
Project Name & Location	Traffic Management System — Coimbatore Urban Study, India
Nature & Collaboration	Academic Project Group Project (Role: Lead Systems Developer)
Level of Study	3rd Year, 6th Semester B.Tech AI & Data Science
Date Carried Out	Academic Year 2024 – 2025
Technical Demonstration	Vehicle Queue Monitoring Algorithms & Dynamic Green Signal Optimization

1. DESIGN CONCEPT & INSPIRATION

Inspiration: Fixed-timer traffic signals cause severe traffic congestion and excess fuel consumption because signal timings do not dynamically adjust to real-time traffic density.

Goals & Objectives: Develop a smart traffic control system that monitors vehicle queue lengths, computes congestion index, and dynamically adjusts green light durations.

2. CONCEPT SKETCHES & DYNAMIC TIMING LOGIC



3. TECHNICAL RESOLUTION & VIABILITY

Resolution: Implemented in Python using data processing algorithms and simulation modules.

Technical Viability: Simulation testing demonstrated a ~28% reduction in average intersection waiting times compared to fixed-timer signal cycles.